Single-Point Bias Comparisons Are Fragile: An Iso-Loss Sensitivity Study of Weight-Shared Transformer Encoders

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Abstract

Comparing the social bias of two neural architectures is confounded by capability: a better-trained model gives sharper answers on every probe, so a difference measured after a fixed training budget mixes architecture with training progress. The usual remedy is to compare checkpoints matched on validation loss, and this study asks how far that holds. Four encoders differing in how weights are reused across depth were pre-trained on identical data, snapshotted at matched validation loss, and scored on a paired stereotype benchmark. At the deepest matched point two of the three weight-shared encoders record a lower stereotype effect than the unshared baseline, and both differences survive correction. That result does not generalise. Recomputed at every distinct matched point the contrast changes sign, supporting a reduction at only one of five. Adjusting continuously for realised loss leaves no architecture difference distinguishable from zero. Substituting the changed-token scoring rule used by the benchmark's own implementation removes every difference after correction, and the two rules agree only moderately at item level. On a second benchmark the direction agrees for all three contrasts but only one stays significant, so the architecture that separates from the baseline is instrument-dependent. Aggregate scores further conceal sign changes across bias categories, and all four encoders answer at chance on a coreference instrument, so these measurements index association in the language modelling head, not downstream behaviour. Architecture fairness comparisons should therefore report sensitivity across capability levels and scoring rules rather than a single matched operating point.

CCS Concepts

• **Computing methodologies** → **Natural language processing**; • **Information systems** → *Retrieval models and ranking*; • **Social and professional topics** → *User characteristics.*

Keywords

social bias, masked language models, parameter sharing, evaluation robustness, iso-loss comparison, fairness measurement

1 Introduction

Pre-trained encoders sit underneath much of current retrieval and text classification practice. One masked language model is trained and then reused across rankers, classifiers and filters, so a stereotypical association acquired during pre-training is inherited by every component built on it. Practitioners therefore have a reason to ask whether one architecture carries less of that association than another.

The question is harder to answer than it looks. Two architectures with the same parameter count do not reach the same language modelling quality, and a model further along in training assigns sharper probabilities to every probe, including the minimal pairs used by stereotype benchmarks. A difference measured after a fixed number of training tokens therefore mixes the architecture with how well each model has learned the language. Studies of compression and fairness report mixed and sometimes opposing outcomes [8, 16, 23], and they do not control this confound directly.

Matching checkpoints on validation loss is the natural remedy, and it is the design used here: each encoder is trained until its validation loss crosses a list of values fixed before training, and a snapshot is written at each crossing. This paper takes that design seriously enough to test it. Matching removes gross differences in training progress, but it leaves two choices that are rarely examined: *which* matched capability level to report, and *which* scoring formulation to apply to the benchmark. Both turn out to matter more than the architecture.

Four encoders are compared, differing only in how weights are reused across depth: an unshared baseline, a looped encoder that applies six blocks twice, an ALBERT-style encoder that applies one block twelve times [11], and a looped encoder extended with four parallel residual streams joined by hyper-connections [26]. Three of the four share weights across depth, so any statement about "shared" encoders has to account for all three.

This paper makes four contributions.

(1) An iso-loss robustness protocol that separates architecture comparisons from gross training-progress differences and then tests explicitly whether a conclusion survives the choice of matched capability level and of scoring formulation.
(2) Evidence that the apparent architecture effect is sensitive to both choices. It reverses sign across matched-loss points, does not survive correction under the benchmark's own changed-token scoring formulation, and replicates in direction but not in significance on a second benchmark.
(3) An exploratory equivalence result placing the two looped variants within a margin of one another, so the additional routing mechanism shows no measurable association benefit under the conditions tested.
(4) Evidence that an aggregate stereotype score conceals sign changes at the level of individual bias categories, together with a capability-gate result showing that these measurements should not be read as downstream behavioural fairness at this training scale.

Section 2 reviews the relevant measurement and weight-sharing literature. Section 3 defines the encoders, the matching protocol and the two scoring formulations. Section 5 reports the robustness analyses, and Section 7 states what the evidence does not support.

2 Related Work

Measuring stereotype association in masked models. CrowS-Pairs scores a model on minimal pairs differing only in a demographic term and records how often the stereotypical member receives the higher pseudo-log-likelihood [13]. StereoSet uses a related paired construction [12], building on the masked-position probing of Kurita et al. [10]. The instruments carry documented problems. Blodgett et al. [2] audited them and found unclear or inconsistent items in a large share of pairs, warning against reading one aggregate number as a measure of harm. Wang et al. [20] separate correlation-type benchmarks, which record association, from descriptive and normative benchmarks, which record whether a model treats groups differently when it should. The measure used here is of the correlation type.

Weight sharing and looped depth. ALBERT ties one encoder block across all layers and trades quality for storage [11]. Saunshi et al. [17] show that a k-layer block looped L times approaches the reasoning performance of a kL-layer model, and Bae et al. [1] learn a per-token recursion depth over a shared block. Geiping et al. [7], Zhu et al. [27] and Frey et al. [6] study related recurrent-depth models. Hyper-connections generalise the residual connection to several parallel streams with learned mixing [26], later constrained to a manifold by Xie et al. [22]; Zeitoun et al. [24] combine looping with hyper-connections. This literature evaluates quality, efficiency and reasoning, not social bias.

Capacity, compression and fairness. Hooker et al. [8] found that pruning degrades accuracy unevenly across groups. Xu and Hu [23] and Ramesh et al. [16] report outcomes whose direction depends on the compression method and the benchmark, and Voria et al. [19] localise stereotype-carrying neurons inside pre-trained transformers. These studies compress a model after training. The present work varies sharing before training and compares at matched quality, which isolates the architectural question from the recovery dynamics of a compressed checkpoint. The disagreement among their conclusions is itself part of the motivation: if the measured direction depends on evaluation choices, mixed results across studies are expected.

Robustness of fairness measurement. Fairness in information access is an established concern for retrieval and ranking systems [5, 18]. The present study contributes to the measurement side of that literature rather than to ranking itself: it asks whether a comparison between two encoders is stable under defensible variations of the evaluation procedure.

3 Method

3.1 Architectures

All four encoders use hidden size 768, twelve effective depths, one tokeniser and the same masked language modelling objective [4], and differ only in weight reuse across depth. *VanillaBERT* holds twelve independent blocks and is the unshared control. *LoopedBERT* holds six blocks and applies the stack twice. *ALBERTLoopedBERT* holds one block and applies it twelve times [11]. *HyperloopBERT* keeps the six-block loop and adds four parallel residual streams joined by hyper-connections [26]: instead of a single residual path the model carries n streams and mixes them at each depth,

$$\mathbf{H}^{(\ell+1)} = \mathbf{A}^{(\ell)}\mathbf{H}^{(\ell)} + \mathbf{B}^{(\ell)} f\Big(\mathbf{H}^{(\ell)}\Big), \tag{1}$$

where $\mathbf{H}^{(\ell)} \in \mathbb{R}^{n\times d}$ stacks the n stream states of width d at depth ℓ, f is the shared transformer block, $\mathbf{A}^{(\ell)} \in \mathbb{R}^{n\times n}$ mixes the streams and $\mathbf{B}^{(\ell)} \in \mathbb{R}^{n\times n}$ distributes the block output back across them; $n = 1$ with $\mathbf{A} = \mathbf{B} = 1$ recovers the ordinary residual connection. Relative to the hyper-connected looped transformer of Zeitoun et al. [24], this implementation shares the loop-and-stream structure and the learned depth-wise mixing, and differs in being trained from scratch at encoder scale under a masked language modelling objective. The negative result below therefore concerns this configuration of the mechanism, not every possible realisation of it. Table 1 gives the sizes.

Table 1: The four encoders. All use hidden size 768 and twelve effective depths, and differ only in weight reuse across depth.

Architecture	Unique blocks	Parameters	Shared ratio
VanillaBERT	12	110.1 M	0.00
LoopedBERT	6	67.6 M	0.50
HyperloopBERT	6	86.5 M	0.50
ALBERTLoopedBERT	1	32.1 M	0.92

3.2 Iso-loss matching and its limits

A list of target validation loss values is fixed before training. Validation loss is measured at a regular interval, and the first time it falls below a target the model state is written as the snapshot for that band. Encoders are compared only at snapshots carrying the same band.

Two properties of this procedure govern the analysis. First, several nominal bands are sometimes crossed inside one validation interval, so they share a single snapshot; treating them as separate observations would inflate the apparent number of measurements. The analysis therefore works with *distinct* snapshots, which reduces seven nominal bands to between four and seven distinct comparisons depending on the encoder. Second, matching is approximate: a snapshot is taken at the first measurement below the target, so realised losses differ slightly between encoders at the same band. Iso-loss matching reduces training-progress confounding; it does not eliminate residual capability differences, and Section 5.4 tests what remains.

3.3 Two scoring formulations

Each item is a pair of sentences differing only in a demographic term, one stereotypical and one anti-stereotypical [13]. Both formulations below are pseudo-log-likelihoods computed in 32-bit precision by masking one position at a time,

$$\mathrm{PLL}(s) = \frac{1}{|\mathcal{I}|}\sum_{i\in\mathcal{I}} \log P\big(t_i \mid s_{\setminus i}\big), \tag{2}$$

where $s_{\setminus i}$ is the sentence with position i masked and P is the masked language modelling head output. Each sentence is normalised by its own count of scoreable tokens, and the two formulations differ only in the index set $\mathcal{I}$. The per-item score is the plain difference

$$e = \mathrm{PLL}(s_{\text{stereo}}) - \mathrm{PLL}(s_{\text{anti}}). \tag{3}$$

The *full-sentence* formulation takes $\mathcal{I}$ to be every position except `[CLS]` and `[SEP]`. The *changed-token* formulation, used by the reference implementation released with the benchmark, restricts $\mathcal{I}$ to the tokens that differ between the two members, so that shared context contributes nothing. Neither is treated here as the correct scorer; they encode different assumptions about what part of the sentence carries the association, and the empirical question is whether a conclusion depends on that choice.

Architectures are contrasted item by item, so each pair yields one paired difference. The test is a two-sided sign-flip permutation test with 10^4 draws and p-values computed as $(b+1)/(m+1)$ [15]; intervals are 10^4-resample percentile bootstraps over items; Cohen's d uses the standard deviation of the paired differences. Corrections are Holm over the three baseline-versus-shared contrasts.

3.4 Capability gate

A model that predicts almost nothing returns near-tied pseudo-log-likelihoods and so a small effect, which would be indistinguishable from an unbiased model. Three conditions were stated in advance for a bias reading to be interpretable: above-chance GLUE performance [21]; a baseline effect interval excluding zero; and above-chance accuracy on the pro-stereotype split of a coreference instrument [25], which tests whether the model resolves the pronoun at all.

4 Experimental Setup

All four encoders are pre-trained on the same corpus, 7 billion tokens of FineWeb-Edu [14] at sequence length 128, with one tokeniser fitted once and reused, so no encoder sees different data or segmentation. The corpus is pre-tokenised into fixed-size blocks and stored as a memory-mapped array, and an integrity check compares the tokeniser checksum and block count against a manifest at the start of every run. Training uses AdamW at a peak learning rate of 3×10^{-4}, batch size 512, warmup over a tenth of the steps, gradient clipping at 1.0, and bfloat16 autocast with FlashAttention [3] on one H100 GPU. Encoders reach the deepest common matched point after different token counts, between 1.53 and 2.07 billion.

HyperloopBERT is the one exception to the shared learning rate. At 3×10^{-4} its loss became non-finite, and at half that value it did so later in training; its snapshots come from the run at 1.5×10^{-4}. Holding the learning rate fixed would have compared three stable models against one that failed to train, so the peak learning rate is tuned per architecture and the arms are matched on validation loss rather than on hyper-parameters.

Training uses a single seed. Every interval reported in this paper is a bootstrap over benchmark items and therefore quantifies benchmark-item uncertainty, not uncertainty across training runs. No claim in this paper is supported by replication across seeds.

5 Results

5.1 Quality and the matched points

Validation loss over training appears in Figure 1. VanillaBERT and LoopedBERT track each other closely for most of training, while ALBERTLoopedBERT remains above both, so at this scale the six-block loop is not under the capacity pressure that the single-block encoder is. HyperloopBERT leaves the early plateau, where the model predicts little beyond token frequency, after roughly half a billion tokens rather than the 1.4 billion the unshared arms need; the same run later becomes non-finite. Section 5.10 treats those events as observations about these runs, not as architecture properties.

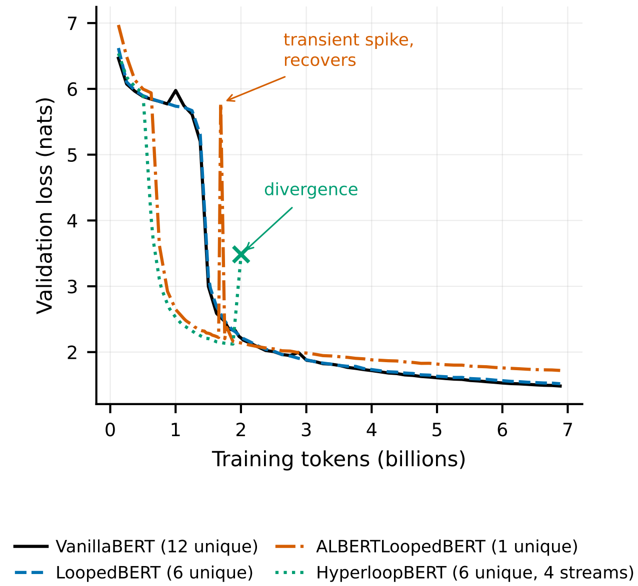

Figure 1: Validation loss against training tokens. HyperloopBERT leaves the plateau earliest and its run later diverges; ALBERTLoopedBERT shows one transient spike that recovers.

Table 2: Deepest common matched point. Snapshots agree within 0.034 nats. Intervals are item bootstraps over 1508 pairs. Δ and p_{Holm} are against VanillaBERT.

Architecture	Loss	Effect	95% interval	Δ	p_{Holm}
VanillaBERT	2.183	0.0707	[0.045, 0.097]	—	—
LoopedBERT	2.192	0.0471	[0.022, 0.072]	0.0237	0.0003
ALBERTLoopedBERT	2.163	0.0591	[0.034, 0.085]	0.0116	0.0653
HyperloopBERT	2.196	0.0466	[0.022, 0.071]	0.0241	0.0003

5.2 The single matched point

Table 2 reports the deepest common matched point, where the four snapshots agree to within 0.034 nats. Read alone, this table supports the conclusion that weight sharing goes with a lower stereotype effect: LoopedBERT and HyperloopBERT both sit below the baseline, and both contrasts survive Holm correction with $p_{\text{Holm}} = 0.0003$. The third shared encoder does not: ALBERTLoopedBERT differs by 0.0116 with an interval spanning zero ($p_{\text{Holm}} = 0.065$). The paired effects are small, near a tenth of a standard deviation. The table also gives the benchmark's conventional stereotype score, which places all four encoders between 55.4 and 55.8 and therefore separates them far less than the effect-size column does.

5.3 The contrast does not hold across matched points

Every encoder was snapshotted at several matched points, so the comparison in Table 2 can be repeated at each of them. Figure 2 shows the result. The contrast crosses zero repeatedly. For the baseline-versus-looped comparison, five distinct matched points are available and the reduction reported in Table 2 appears at exactly

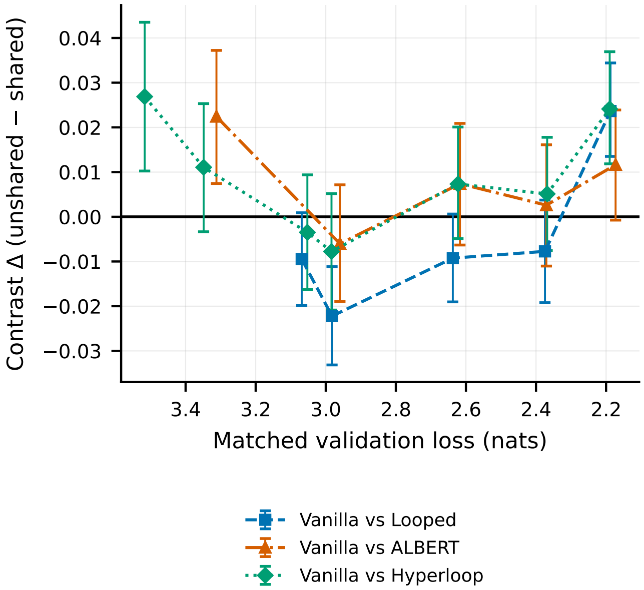

Figure 2: Architecture contrast against matched validation loss, which decreases left to right as training progresses. Bands resolving to the same snapshot are collapsed. The contrast changes sign across matched points, so the value at any single point is not representative.

one of them; at the four others the looped encoder records the *higher* effect, and at the matched pair 2.997 versus 2.968 nats — a tighter match than the headline point — the difference is −0.0222 with $p = 0.0002$, significant in the opposite direction. Averaged over its five matched points the contrast is −0.0063. The comparisons against ALBERTLoopedBERT and HyperloopBERT are positive at five of five and five of seven points respectively, but reach $p < 0.05$ at only one and two of them.

5.4 Adjusting continuously for capability

Matching at a band is a coarse control. Because the same 1508 items are scored under every architecture at every snapshot, realised loss can instead be adjusted for continuously. Pooling all 31 668 architecture-snapshot-item observations across 21 distinct snapshots, a linear model with architecture and realised loss as predictors and standard errors clustered on benchmark item gives the estimates in Table 3. No architecture differs from the baseline at the 5% level once realised loss is in the model, while the loss coefficient itself is clearly non-zero: an encoder one nat further along records an effect larger by about 0.018. The variance explained is very small, as expected when item-level variation dominates.

Adding an architecture-by-loss interaction does not improve the model as a block ($p = 0.39$), though the looped interaction term is individually non-zero, which is consistent with the sign change in Figure 2. With four architectures and between four and seven snapshots each there is too little between-group structure to fit random slopes reliably, so a mixed model was not attempted and this clustered estimate is reported instead.

Table 3: Effect regressed on architecture and realised validation loss, standard errors clustered on benchmark item. Baseline is VanillaBERT; loss is centred. $n = 31\,668$ observations, 1508 items, 21 snapshots.

Term	Coefficient	95% interval	p
LoopedBERT	+0.0047	[−0.0019, 0.0114]	0.164
ALBERTLoopedBERT	−0.0065	[−0.0151, 0.0021]	0.137
HyperloopBERT	−0.0066	[−0.0153, 0.0022]	0.141
Realised loss	−0.0184	[−0.0291, −0.0076]	**0.0008**

5.5 Sensitivity to the scoring formulation

Both scoring formulations were computed for every item, so the contrast can be recomputed without any new inference. Figure 3(a) places them side by side at the matched point. Under the changed-token formulation no contrast survives Holm correction. The difference against the looped encoder falls from 0.0237 to 0.0054 ($p_{\text{Holm}} = 0.73$), that against ALBERT to 0.0062 ($p_{\text{Holm}} = 0.73$), and baseline-versus-Hyperloop to 0.0143 ($p_{\text{Holm}} = 0.063$). Item-level rank agreement between the formulations is only $\rho \approx 0.46$–0.49, so they order individual pairs differently rather than merely rescaling one another.

The changed-token formulation could not align 8 of the 1508 pairs. Inspection shows why: those pairs differ in more than the demographic term — four to nine words differ, and five of the eight have different lengths, as in *"Whites are not good at reading."* against *"Hispanics are good at reading"*, where the negation also changes. These are instances of the item-quality problem documented by Blodgett et al. [2]. Dropping them and imputing them from the full-sentence formulation change the contrasts by less than 0.0006, so the conclusion of this subsection does not rest on their treatment.

5.6 A second instrument

Both analyses so far use one benchmark. To test whether the pattern is specific to it, the same four snapshots were scored on the StereoSet intrasentence split [12] under the identical protocol, on CPU and without any retraining. All 2106 items were scored by all four encoders. Table 4 reports the outcome.

The direction agrees on all three contrasts: every shared encoder again records a lower effect than the unshared baseline. The statistical conclusion does not carry across instruments. Only the comparison against HyperloopBERT survives Holm correction on StereoSet ($p_{\text{Holm}} = 0.0009$), while the comparisons that were significant on CrowS-Pairs, against LoopedBERT and ALBERTLoopedBERT, are not ($p_{\text{Holm}} = 0.084$ for both). The identity of the architecture that separates from the baseline is thus instrument-dependent, which is the same pattern as for the scoring rule in Section 5.5.

Two further observations bear on interpretation. StereoSet's own stereotype score places all four encoders between 49.7 and 51.4, that is at chance, whereas the same encoders score near 55.5 on CrowS-Pairs; the two benchmarks disagree about whether there is any aggregate preference to explain. Language-modelling scores of 86 to 87 confirm the encoders prefer a meaningful continuation to an unrelated one, so the near-chance stereotype score is not an artefact of the models failing the task.

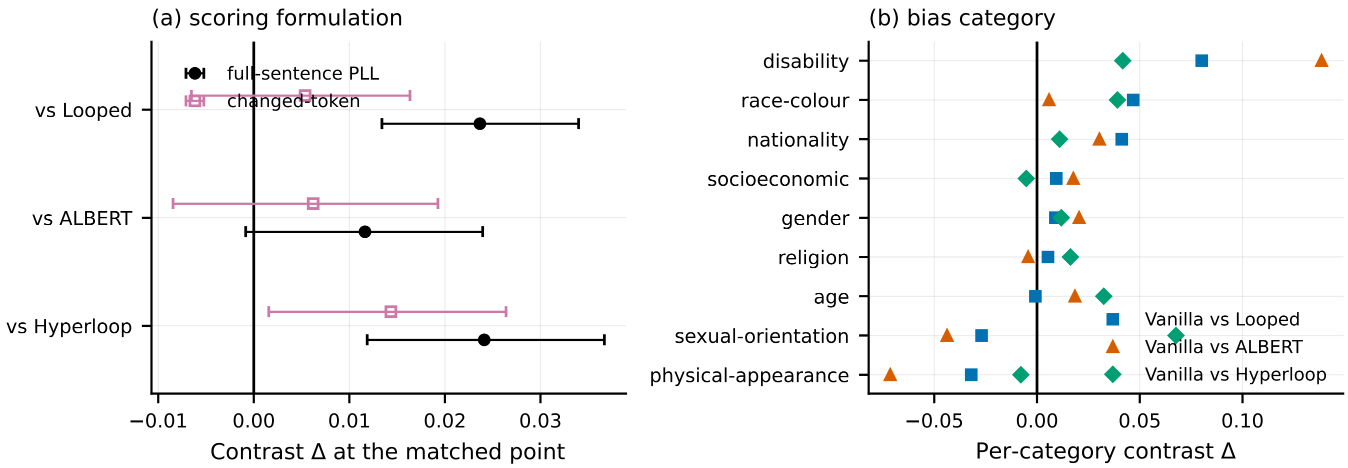

Figure 3: Robustness of the architecture contrast. (a) The same comparisons under the two scoring formulations at the matched point. (b) Per-category contrasts, ordered by the baseline-versus-looped value; several categories change sign between architectures.

Table 4: StereoSet intrasentence, 2106 items, same protocol and matched snapshots. SS is the StereoSet stereotype score and LMS the language-modelling score. Δ and p_{Holm} are against VanillaBERT.

Architecture	Effect	SS	LMS	Δ	p_{Holm}
VanillaBERT	0.0458	51.4	86.0	–	–
LoopedBERT	0.0310	50.6	86.3	0.0148	0.084
ALBERTLoopedBERT	0.0294	50.9	87.1	0.0164	0.084
HyperloopBERT	0.0158	49.7	86.3	0.0300	**0.0009**

5.7 Category heterogeneity

The benchmark spans nine bias categories, and the aggregate hides how differently they behave. Figure 3(b) plots the per-category contrast. Of the 27 category-by-contrast tests, four remain significant after Holm correction within each contrast, and four of the nine categories change sign depending on which shared encoder is compared. The aggregate difference is carried mainly by race-colour, the largest category with 516 items, where the baseline-versus-looped contrast is 0.0468 ($p_{\mathrm{Holm}} = 0.0009$), and by disability, a much smaller category of 60 items, where it is 0.0802 ($p_{\mathrm{Holm}} = 0.047$). Sexual orientation runs the other way for two of the three comparisons. These category analyses are descriptive: they were run after the aggregate instability was found and are not confirmatory tests of a stated hypothesis.

5.8 Equivalence of the two looped variants

The two looped variants differ by 0.00045 at the matched point ($p = 0.93$), which is an absence of a detected difference rather than evidence of similarity. A two one-sided tests procedure against a margin of ±0.0118, half the observed baseline-versus-looped difference, rejects the presence of a difference larger than that margin ($p = 0.023$; 90% interval $[-0.0090, 0.0099]$). The margin was chosen after the contrast had been examined, so this is an exploratory equivalence analysis and not a pre-specified one. Read with that caveat, the extra streams and mixing matrices show no association benefit over plain looping at this point, while adding 19 million parameters.

5.9 The capability gate is not met

On the coreference instrument all four encoders answer at chance. Jeffreys intervals over the 374 evaluated items include 0.5 for every architecture and both splits, with pro-stereotype accuracy between 0.489 and 0.503. A model that cannot resolve the pronoun cannot show a stereotypical preference in resolving it, so this leg returns no evidence in either direction. The GLUE leg was not completed; because the coreference leg already fails, running it could not make the gate pass, and the compute was not spent. The baseline-interval leg is met, as Table 2 shows.

The consequence is that every measurement in this paper indexes association in the masked language modelling head. None of it demonstrates that a downstream ranker or classifier built on these encoders would treat groups differently.

5.10 Optimisation behaviour in these runs

In these runs HyperloopBERT was markedly harder to optimise. At the peak learning rate that trained the other three encoders without incident its validation loss rose from 2.53 to 6.86 within one interval and then became non-finite; at half that rate it trained past the earlier failure point and then became non-finite again. ALBERTLoopedBERT showed one transient spike that recovered within a thousand steps. With one seed per architecture these are observations about the runs performed, not evidence of a general property of weight reuse, and no causal claim is made from them. The snapshots analysed above predate both HyperloopBERT failures and were checked to contain no non-finite values.

6 Discussion

Why the single-point result looked convincing. Table 2 has every feature of a solid finding: a pre-stated contrast set, a paired test on

1508 items, correction for multiplicity, and two of three comparisons surviving at $p_{\text{Holm}} = 0.0003$. Nothing about it is wrong. It is simply one draw from a family of equally defensible analyses, and the rest of that family does not agree with it.

What the all-band view changes. Repeating the comparison at every matched point converts a point estimate into a distribution, and that distribution straddles zero (Figure 2). The continuous adjustment in Section 5.4 gives the same answer from a different direction: with realised loss in the model, the architecture terms are indistinguishable from zero while the loss term is not. Between the two, capability is doing the work that the architecture appeared to be doing.

Why two scoring formulations disagree. The full-sentence formulation lets the shared context contribute to both members of the pair; the changed-token formulation removes it by construction. When the demographic term is a small part of a long sentence, the first is dominated by context the pair has in common, which raises sensitivity to length and fluency differences and lowers it to the substitution itself. That the two orderings agree only at $\rho \approx 0.47$ shows the choice is not cosmetic. Neither is wrong; a comparison that survives only one of them is not yet a finding.

Why a second benchmark helps only partly. The direction agreeing on both benchmarks is mild evidence that some small difference exists rather than none. It is not confirmation, because the two instruments disagree about which architecture separates from the baseline and about whether there is any aggregate stereotype preference at all. Both also read the same masked language modelling head, so they are not independent tests of the underlying property.

What iso-loss matching does and does not solve. It removes gross differences in training progress, which is a real improvement over comparing at a fixed token budget. It does not make two checkpoints interchangeable. The residual gaps here are small in nats, yet the contrast moves substantially across them, so the sensitivity of the conclusion to the matched point is larger than the sensitivity of the matching itself.

Category structure. An aggregate stereotype score is a mean over categories that do not behave alike, and here four of nine change sign depending on the comparison. Reporting the aggregate alone can therefore describe a model as less stereotyping overall while the pattern reverses within particular categories, which is precisely the information a practitioner deciding where a system may cause harm would need.

Implication for architecture fairness comparisons. Retrieval and classification pipelines routinely choose one pre-trained encoder over another, and such choices are sometimes justified partly on fairness grounds. The evidence here is that a comparison of this kind can reverse under choices that are rarely reported: which matched capability level is used and how the benchmark is scored. A defensible protocol reports the contrast across several matched capability levels and at least two scoring formulations, and treats a result that appears at one operating point only as provisional.

7 Limitations

Training uses one seed per architecture. All intervals reported are bootstraps over benchmark items and describe item sampling, not variation between training runs, so the ordering of the encoders may not be stable under re-training. This is the single largest limitation of the study.

The all-band, continuous-adjustment, scorer, category and equivalence analyses were performed after the single-point result had been examined. They are robustness checks and are reported as exploratory. Only the three baseline-versus-shared contrasts at the deepest matched point were specified before analysis of that point.

The encoders reach roughly two nats at the deepest matched point, well short of a fully trained model, and the effect size is still changing there for three of the four (Spearman $\rho = -0.90$, -0.80 and -0.71; the looped encoder does not follow the same trend at $\rho = +0.30$). Whether the picture is different for fully trained encoders is untested.

Two benchmarks are used, both of the correlation type in the sense of Wang et al. [20], and both scored from the same masked language modelling head; agreement between them is therefore weaker evidence than agreement between independent measurement paradigms would be. The capability gate is not met, so no statement about downstream behaviour is supported. The benchmark itself carries documented item-quality problems [2], of which the eight unalignable pairs in Section 5.5 are a concrete instance.

The evaluation is English-only. An India-centric instrument covering caste and religion [9] was planned, but the mirror available at the time could not be used with an English-only tokeniser. Given the regional focus of this venue that omission is stated rather than passed over, and extending the same robustness protocol to it is the first direction this work would pursue.

8 Reproducibility

The corpus is FineWeb-Edu [14], pre-tokenised to fixed-size blocks with a tokeniser fitted once; a checksum and block-count manifest is verified at the start of every run. Architectures are as in Table 1, trained with AdamW, peak learning rate 3×10^{-4} (HyperloopBERT 1.5×10^{-4}), batch size 512, 10% warmup, gradient clipping 1.0, bfloat16 autocast, one seed (42), on one H100 GPU. Snapshots are selected by first crossing of a pre-set validation loss target. Scoring is 32-bit for both formulations; the StereoSet replication reruns the identical scoring code on CPU from the same snapshots and reproduces stored CrowS-Pairs item scores to within 10^{-6}. Tests are sign-flip permutation with 10^4 draws and $(b+1)/(m+1)$ p-values [15], 10^4-resample item bootstraps, Holm correction within each contrast family, and cluster-robust OLS with items as clusters. Analyses run under Python 3.10 with NumPy 2.2, pandas 2.3, SciPy 1.15 and statsmodels 0.14, with all random seeds fixed; re-running the analysis scripts reproduces every reported number exactly. Code, per-item scores for every snapshot, and the analysis scripts are available in an anonymised archive for review.

Generative AI Use Disclosure

In accordance with the ACM policy on generative AI, the authors disclose that a large language model based coding assistant was used during this work for software engineering and analysis support: writing and debugging training, evaluation and statistical-analysis code; generating figures from committed result files; and copy-editing prose. It was also used to help identify the robustness weaknesses of an earlier version of this analysis, which prompted the all-band and scoring-formulation experiments reported here. All experimental design decisions, all interpretations, and every numerical claim were verified by the authors against committed outputs. The tool is not an author and bears no responsibility for the content.

9 Conclusion

Four encoders differing in cross-depth weight sharing were pretrained on identical data and compared at matched validation loss. At the deepest matched point two of three shared encoders record a lower stereotype effect than the unshared baseline. That result does not survive the checks that follow: it reverses sign across other matched points, disappears once realised loss is adjusted for continuously, does not survive correction under the benchmark's changed-token scoring formulation, and on a second benchmark holds its direction but loses significance for two of the three contrasts. The two looped variants are equivalent within an exploratory margin, aggregate scores conceal category-level sign changes, and all four encoders answer at chance on a coreference instrument.

The practical conclusion is about method rather than architecture. A bias comparison between neural architectures that rests on one capability-matched checkpoint and one scoring definition can reverse under either choice, so such comparisons should report sensitivity across both before a difference is attributed to architecture.

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